

Problem-Solution Fit:

Date	2 July 2026
Team Id	XXXXX
Project Name	Opti Crop : Smart Agricultural Product Optimization Engine
Maximum marks	5 marks

Problem-Solution Fit:

1. CUSTOMER SEGMENTS (CS) - Farmers (Small & Large Scale) - Agricultural Researchers - Agribusiness Companies - Government Agricultural Officers	6. CUSTOMER LIMITATIONS (CL) - Limited internet access - Lack of technical knowledge - Budget constraints - Limited access to agricultural experts	5. AVAILABLE SOLUTIONS (AS) Solutions: Traditional farming, agricultural experts, farming websites. Pros: - Easy to access - Expert guidance Cons: - Time-consuming - Costly - Less accurate than AI predictions
2. PROBLEMS / PAINS (PR) - Low crops yield due to improper crop selection. - Lack of real-time soil and weather analysis. - High fertilizer and irrigation costs. - Unpredictable weather affecting crops.	9. PROBLEM ROOT / CAUSE (RC) - Incorrect crop selection. - Climate change. - Poor soil nutrient management. - Lack of scientific farming practices.	7. BEHAVIOR (BH) - Consults local farmers or experts. - Searches online for crop information. - Uses previous farming experience. - Performs soil testing before sowing.
3. TRIGGERS TO ACT (TA) - New farming season begins. - Soil test results are available. - Weather conditions change. - Need to improve productivity	10. YOUR SOLUTION (SL) Smart Agricultural Production Optimization Engine is an AI-based crop recommendation system that analyses soil nutrients, temperature, humidity, rainfall, and pH to recommend the most suitable crop, helping farmers improve	8. CHANNELS OF BEHAVIOR (CH) ONLINE - Agricultural websites - Government portals - YouTube - Mobile apps

	productivity and make data-driven decisions.	OFFLINE • Agricultural officers • Local markets • Farmer meetings • Training programs
4. EMOTIONS Before: Confused, worried, uncertain about crop selection. After: Confident, satisfied, optimistic about higher yields.		